OpenClaw Scheduled Tasks Tutorial

> Before starting, please complete the [01-Hardware Configuration], [02-openclaw API-KEY Configuration], [03-openclaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solution Configuration Tests] under the `Prerequisites` directory. If you are currently only using TUI / whatsapp, you can skip `04` for now. This article assumes the basic environment has been prepared and only expands on "fixed-time automatic script execution" part of the content.

1. Tutorial Objectives

This article demonstrates through "AI-generated script + Linux scheduled execution" how OpenClaw upgrades from immediate dialogue to automated peripheral applications. After completing this tutorial, readers can:

- Understand the basic implementation approach of OpenClaw scheduled tasks
- Distinguish the scenarios suitable for `HEARTBEAT` and `cron` respectively
- Describe automation requirements through three methods: whatsapp, TUI, and voice code

2. Hardware Preparation

- OpenClaw mainboard
- DHT temperature and humidity sensor
- OLED display
- RGB light strip / RGB light bead module
- `[Optional] CSI camera`
- `[Optional] AI voice module + speaker`

Description:

- The core of this tutorial is "scheduled execution", not requiring all peripherals to be connected simultaneously
- Connect the corresponding hardware first for the type of automation task you want to do

3. Test Scheduled Task Basic Capability

Before starting this project, please complete one `openclaw tui` basic dialogue self-check; if you haven't done it yet, see [05-Three Dialogue Solution Configuration Tests]. After passing, enter scheduled task specific testing.

It is recommended to test in the following order:

1. First enter OpenClaw TUI
2. Let OpenClaw generate a simple script
3. Run the script manually in terminal, confirm functionality is normal

Terminal input:

```
openclaw tui
```

It is recommended to first try the following type of requirements:

- Help me write a script to display current temperature and humidity on OLED, save to desktop
- Generate a script that reads temperature and humidity and prints results every time it runs
- Write a script to make RGB light green for 3 seconds then turn off

After the script is generated, first run it manually, for example:

```
python3 /home/jetson/Desktop/your_script.py
```

4. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Positioning
whatsapp	Remote description of automation requirements	Suitable for first clarifying "what to do on schedule"	Used to generate scripts and confirm requirements
TUI	Local creation and debugging of scheduled tasks	Easiest to jointly debug scripts and `cron`	Preferred implementation solution
Voice code	Voice description of automation scenarios	Closer to desktop assistant experience	Suitable for demonstrating "one sentence generates automation"

If optional AI voice module + speaker are selected, whatsapp dialogue, tui, whatsapp dialogue need to broadcast OpenClaw's reply content. You can run the following program in terminal. AI voice module dialogue does not need to run repeatedly

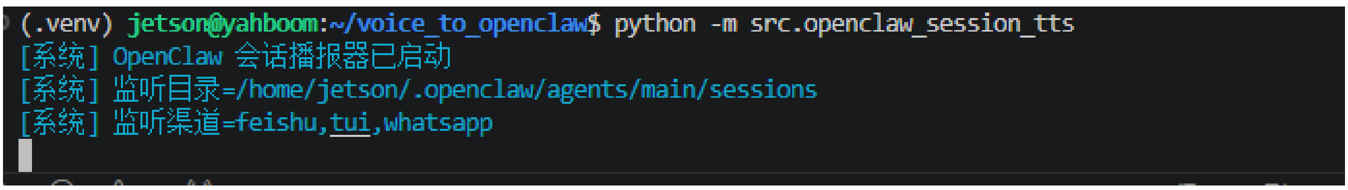
- Modify whether to enable broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  #This parameter controls whether to broadcast. After modification and saving, run the following program in terminal
```

- Terminal input

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```



4.1 whatsapp Solution

For whatsapp connection steps, please directly refer to [05-Three Dialogue Solution Configuration Tests]. This section is more suitable for describing requirements, for example:

- Help me create a scheduled task: make RGB lights up red every minute as a remind, then turn off the light.

20:42

+86 153 3885 7526 (你)
给自己发消息

Help me create a schedule task: make RGB lights up red every minute as a reminder, the turn off the light.

20:41

Schedule task created successfully!

Task Details:

- **Name:** RGB Red Minute Reminder
- **Schedule:** Every 60 seconds (1 minute)
- **Job ID:** `e13c213c-223e-4408-92f5-ef708d8d41ce`
- **Status:** Enabled ✅
- **Next Run:** In ~60 seconds

What happens each minute:

1. 💡 RGB lights turn **RED** at 80% brightness
2. ⏱️ Stay lit for 2 seconds as a reminder
3. ⛔ Lights turn off automatically

You'll receive a WhatsApp notification when the task runs. The red minute reminder is now active! ⏰🔴

20:42

RGB minute reminder executed successfully:

- Turned on all RGB LEDs (blue colo 80% brightness)

4.2 TUI Solution

If you have completed the TUI self-check in Chapter 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then you can directly enter project-related commands, for example:

- Help me create a scheduled task: read temperature and humidity every minute and display on OLED

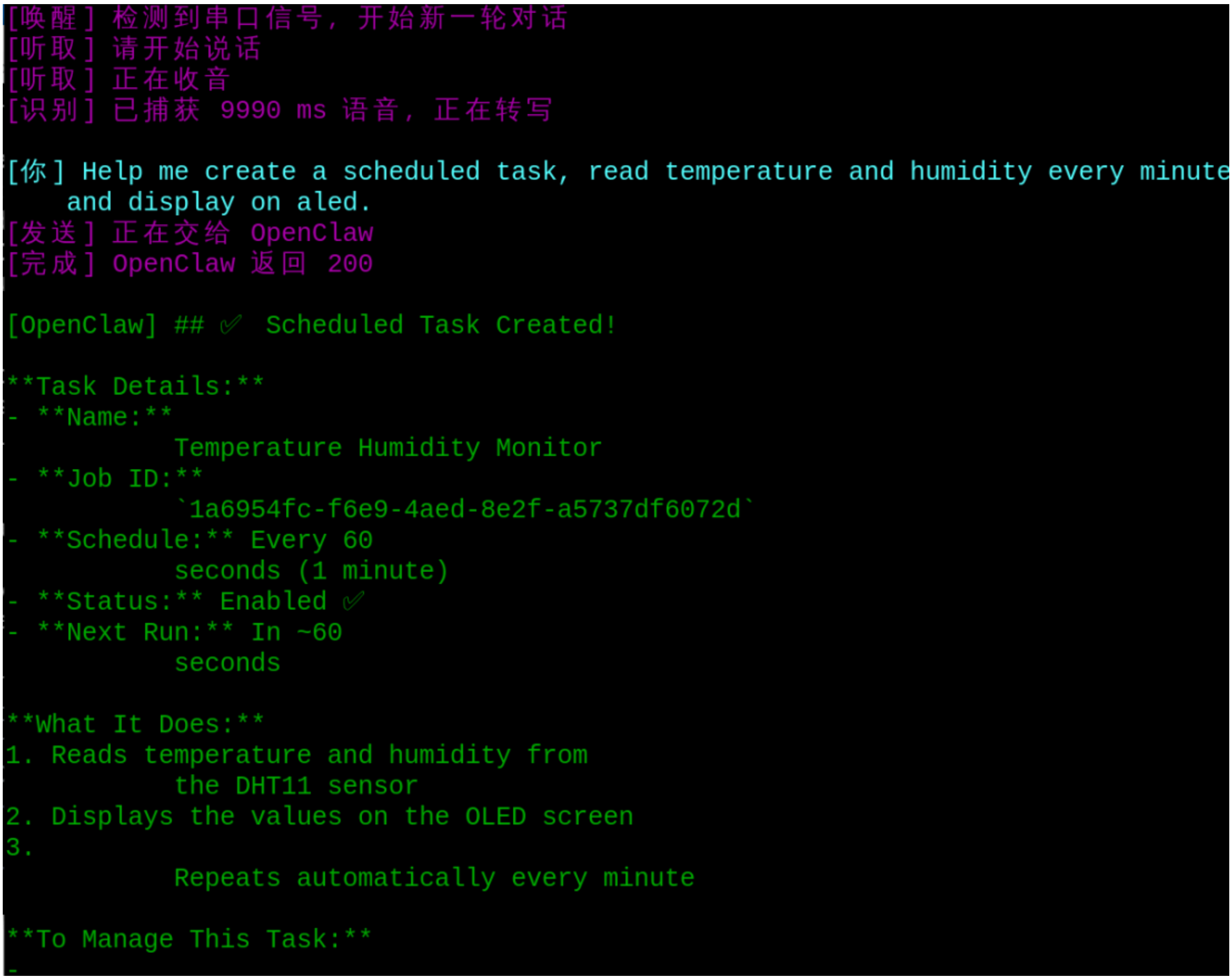
```
Help me create a scheduled task: read temperature and humidity every minute
and display on OLED

Scheduled task created successfully!

Task Details:
- Name: Temperature Humidity Dashboard
- Schedule: Every 1 minute (60,000ms)
- Job ID: a43df4b4-6c47-464e-8cce-95ed29bc8a36
- Status: Enabled ✅
- Next Run: In ~60 seconds

What it does:
- Reads temperature and humidity from the DHT11 sensor
- Displays the values on the OLED screen
- Runs automatically every minute in the main session

To manage this task:
- View all jobs: Use cron list command
- Remove job: Use cron remove with the job ID above
- Pause job: Use cron update to set enabled=false

The temperature dashboard will now update automatically every minute!
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
83k/1.0m (8%)
```

4.3 Voice Code Solution

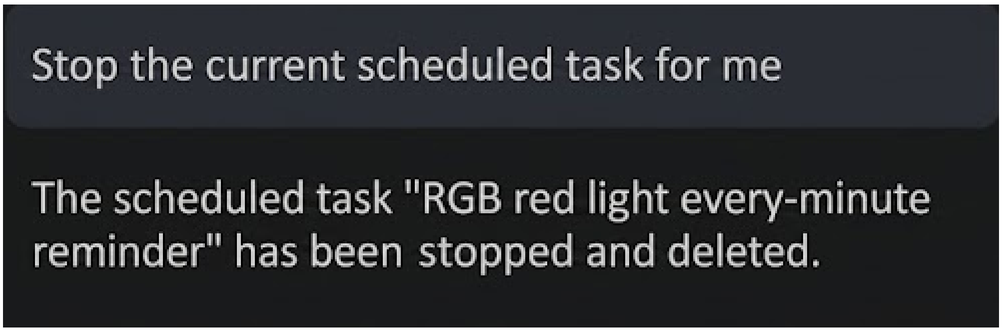
```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [04-AI Voice Interaction Configuration]. After waking up, you can directly say control content, for example:

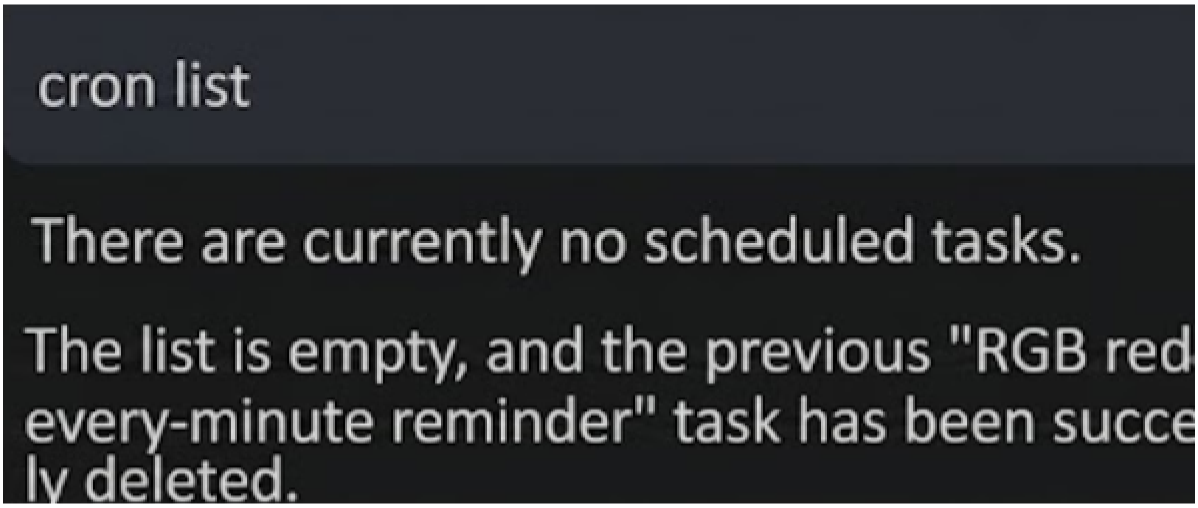
- Help me create a scheduled task: read temperature and humidity every minute and display on OLED



- If you want to stop the scheduled task, you can say in the current dialogue session: stop the current scheduled task for me



- You can use the command `cron list` to view current scheduled tasks



5. Project Implementation

Note: The following are all dialogue demonstrations using `openclaw tui` method. You can also choose whatsapp or voice method to complete.

5.1 Scenario One: Hourly Temperature and Humidity Sampling

Experimental Objective

Let the device automatically read temperature and humidity once per hour, and record results or synchronize to screen.

Effect Description

This is the most basic and common scheduled task scenario. It can later be further expanded into logs, dashboards, environmental alerts, or plant care projects.

Example Instructions

- Help me make a scheduled task: read temperature and humidity every minute and display on OLED

5.2 Scenario Two: Timed Status Reminder

Experimental Objective

At a fixed time, use RGB or OLED to make a clear reminder.

Effect Description

This type of task is suitable for water drinking reminders, class reminders, patrol reminders, status switching reminders. It does not emphasize continuous running, but emphasizes triggering a clear action at a certain time point.

Example Instructions

- Help me make a scheduled task: make RGB light red every minute to remind, then turn off

5.3 Scenario Three: Timed Patrol Script

Experimental Objective

String multiple peripheral actions together to form a periodically executable small automation task.

Effect Description

For example, read temperature and humidity once per hour. If status is abnormal, light alert, and display results to OLED. This type of task is already close to a complete small project, and is also the most obvious place where "AI-generated script" combines with "scheduled execution".

Example Instructions

- Help me make a scheduled task: read temperature and humidity once per minute, light red when abnormal, and prompt on OLED